

Introduction: The history of the CubeSat by Bob Twiggs and Jordi Puig-Suari

1 The CubeSat standard

The CubeSat came about from a project at the Aeronautics and Astronautics Department that Stanford University had with DARPA and the Aerospace Corporation in 1998. The Aerospace Corporation wanted to launch a little satellite (picosat) the size of Klondike ice cream bar as part of a Defense Advanced Research Projects Agency (DARPA) program. The graduate engineering students at Stanford that had been working on microsats since 1995 decided that it would be challenging to build a launcher for this little picosatellite. They designed a deployer that fit inside one of our microsattellites. The launcher was built like the cartridge holder for a gun. The cartridges were pushed down into the holder where a collapsed spring is ejected when needed. The students designed and built a rectangular device that would hold the picosats. The microsat was called Orbiting Picosat Automated Launchers (OPAL) and with the mission to deploy the picosats when in Earth orbit. OPAL and the picosats were launched in 2000. The little picosats from DARPA worked as planned. OPAL and the picosats demonstrated the feasibility for students to work on smaller satellites like picosats. With the existing student microsatellite program, it was difficult to get the students to finish a design because there was too much room and they kept adding to the satellite experiments.

After the OPAL launch, we were working on a project with Jordi Puig-Suari at Cal Poly at San Louis Obispo, CA. We had an opportunity to launch a small satellite on a Russian rocket if we could design a very small satellite like the picosats. It turned out that launch opportunity was canceled. We then started thinking about picosat-sized satellites as a standard. The idea that evolved from this led to the CubeSat. The design challenge was to make something like this little picosat that was like the size of an ice cream bar and a launcher that could launch several of them at once. But, to make them useful, we needed to put more solar cells on them than could only be mounted on the flat sides of the picosat. That need led to the thought of making the picosat in a cube shape. Using some solar cells that were donated from JPL, we tried these 2×4 cm cells to see how many solar cells could be put on the square surface like a cube and still have a method of holding the cube in the launcher. Laying these cells out to get a reasonable voltage string took about a square of 4 in. To find a model, we went to the local plastic shop where a 4-in. cube was selected. At that time, beanie babies were the rage, so the cube selected was a display case for the beanie babies.

We purchased several 4-in. beanie baby cubes and began to design a way to contain them in a deployer. The method of holding the picosats was by the corners that were chamfered. In the picosat, they were secured in the launcher only by the force of the launcher door. It had worked for the picosats, so this same method was used with the CubeSat deployer. Being held by the corners with 10 one-thousands clearance, it would not move significantly during vibration. A quarter inch was allowed on the edges for the rails. To have multiple CubeSats in a launcher, it would require some way to separate the cubes. We bought quarter inch plastic cubes that were put on the ends of the 4-in. cubes to keep the adjacent cube faces from contacting. These were later called separation feet.

We were using the standard English units in our engineering programs. The aerospace industry uses metric units so we decided that the students should learn metrics. The 4-in. cube is roughly equivalent to 10 cm, so even though the model was based on a 4-in. cube, we changed the design to a 10-cm cube. At that time in the late 1990s, Lockheed and Martin Marietta had built a satellite, the Mars Climate Orbiter that went to Mars and it failed. The reason it failed was because one team was working in English units and one was working in metric units. That was a Mars mission that burned up in the Martian atmosphere because of a mix up. We decided that US engineering students should learn the metric system and changed the CubeSat standard to 10cm. The accepted aerospace definition for a picosat was 1 kg. If a 10-cm cube is filled with water, it weighs a kilogram. The mass allocation became 1 kg. That was how the CubeSat standard was determined.

2 The PPOD

After the initial model cube was selected, I built a plastic deployer that held three CubeSats with the rails and that included feet on the CubeSats. This model was given to Jordi Puig Suari at Cal Poly. They had a reputation as an innovative mechanical engineering program and I believed that they could come up with a deployer design for the CubeSats. Jordi's students came up with all sorts of ideas and finally settled on the one that I had sent to them based on the OPAL design, basically a Jack-in-the-Box. Cal Poly then wrote specs that were the key factor in making the CubeSat form factor successful in being adopted by the space industry. The deployer was named the Poly Picosat Orbiting Deployer (PPOD).

The size of the PPOD was defined by the only secondary launch accommodation specification available at the time: the Delta II secondary payload specification CubeSats. The dimensions of that location are the reason for the PPOD holding three 1 U CubeSat. It is interesting that CubeSats were never launched using that Delta II specification, even when they were launched later on the Delta II rocket.

The key function of the PPOD was to protect the launch vehicle and the primary payload from any mechanical failure of the CubeSats. This eliminated one of the main concerns that launch providers had about including university payloads in their launches. As a result, the PPOD was an enclosed box with very strong walls, not a traditional mass optimized space system.

3 The first CubeSat launches

Cal Poly took on the task as the integrator for launching the first CubeSat. At that time the Russians were the only people that would even talk about launching these things because they needed the money. The cost for first launch was \$30,000 for a 1U CubeSat. There were several PPODs with CubeSats that were launched on Russian rockets that comprised the early CubeSat launches.

The first group of CubeSats was launched in June of 2003 from Plesetsk, Russia, on a Eurokot, utilizing the Russian Multiple Mission Orbit Service. Several CubeSats were put into a Sun-synchronous orbit. These were the Danish AAU CubeSat and DTUSat, the Japanese XI-IV and CUTE-1, the Canadian Can X-1, and our Quakesat. QuakeSat was a collaboration between Stanford and a small company in Palo Alto called Quake Finder. They were looking for earthquakes, and they were looking for the energy that is emitted from an earthquake right before they occur. They had a number of low-frequency ground sensors that they had put out around California. With these sensors, they found some evidence that before the Loma Prieta earthquake in the Bay Area there were definite low-frequency signals emitted. They built this satellite with a pop-out low-frequency antenna, and we flew it on the Russian rocket. After the fact, they picked up some signals from an earthquake that was near Lompoke in California. There was a satellite built by the Danish Technical University (DTUSat) and a Japanese satellite from the University of Tokyo (CUTE-1). CUTE-1 had a camera on board. We got the CubeSats launched and were successful. The QuakeSat group put a ground station up in Alaska to make contact with their CubeSat. Several of these first missions were very successful.

It should be noted that QuakeSat was the first 3U CubeSat and opened the door for the most popular form factor in the CubeSat world and required the modification of the standard after the first launch.

The Japanese students asked me if a camera would work on a CubeSat. I said, well I don't know if they could get an image that shows you very much. They went ahead and built it anyway. After it was launched in 2003, the students started sending me pictures that the satellite had taken. For years, they kept sending me pictures. The Japanese CubeSat, CUTE-1, lasted for years, becoming the longest-lived CubeSat, and may still be working. This first group of CubeSats that was launched was very successful.

The next Cubesat launch did not happen until July 26, 2006. Again a Russian vehicle, in this case a Dnepr, was selected since Russia was the only country willing to fly CubeSats. The launch included 14 CubeSats from 11 organizations from around the world. Unfortunately the launch was a failure, and all the CubeSats were lost. This failure had a surprising result since the space community rallied around the students who saw their hard work crash into a field in Kazakhstan. The efforts to find US launch opportunities for university CubeSats increased dramatically, and after a second Dnepr launch in 2007, the Cal Poly team changed their focus to US launch vehicles, and within a few years had successfully launched CubeSats on most major US launch vehicles.

4 CanSat—The proto-history of the CubeSat

In 1998 at a joint US and Japanese student conference, I proposed the idea of launching a spacecraft the size of a soda (Coke) can into space. This initial idea would later evolve into other nanosatellite projects and influenced the idea for the CubeSat. The idea at that time was to take a Coke can, put some electronics in it, put it on a high-power amateur rocket, fly it, and eject it out on a parachute. Professor Shinichi Nakasuka and his students from the University of Tokyo began developing the CanSat. There is now an annual event with the high amateur rocket Aero Pac group in Northern California going to the Black Rock desert in Nevada in September every year to launch CanSats and rovers.

4.1 *Use of amateur radio frequencies*

When we proposed the CubeSat, we wanted to use the amateur radio frequencies, and the AMSAT guys were not in favor of us using these amateur radio frequencies. The AMSAT guys had launched several successful amateur satellites, and they had a table of the frequencies allocated to them, so I asked them if they would give me a frequency, and they said we were not qualified to use these frequencies, even though we had the proper amateur licenses.

Because of the years of experience of AMSAT members building and launching small satellites, we were told that we did not have the technical capability to build satellites. They had an Internet blog at the time where they put some less than favorable comments on new proposals for the use of space like their program. There were unfavorable comments about student university projects using those frequencies. It was, however, legal for other licensed amateur to use them. So we then picked a frequency band to use at Stanford and suggested to other university CubeSat developers the frequencies they could use. We all then applied to the FCC and got them approved, without the blessing of the AMSAT group. It took several years for the AMSAT international group to help coordinate amateur frequencies for University satellites and to accept the concept of the CubeSat.

5 NASA and NSF get in the game

Initially the scientific side of the aerospace community and NASA paid no attention to the CubeSats. In 2008, that changed, when Terese Jorgensen at the National Science Foundation organized a CubeSat conference to explore the possibility of using CubeSats for atmospheric research. Dr. Jorgensen saw the potential and convinced NSF to offer contracts to support the first CubeSat science missions.

I caught Dr. Jorgensen right at the end of the conference to talk to because I thought it was ironic that she had not become interested in CubeSats before 2008, 5 years after the first launch. I asked her why NSF waited so long since we announced this initiative in 2003. Dr. Jorgensen said that NSF was waiting to see if it worked before committing. Dr. Jorgensen went to NASA before committing to sponsor the conference, since

satellites in the past had been managed by NASA. When Dr. Jorgensen contacted one of Chief Directors at NASA that builds satellites and asked, “should NASA build CubeSats?”. He said, “we build real satellites like Ferraris; we don’t build those Hot Wheel type satellites.” But once NSF committed to CubeSat programs, other aerospace companies started to consider them.

I had predicted that CubeSats would have an exponential growth with the students building them and that the innovation would really grow when there was an acceptance of failure if the students had built them. In my opinion, NASA had become extremely conservative, building bigger and more expensive satellites. Failure was not an option for NASA. It is hard to innovate if you don’t take chances. I thought that the CubeSat could be a stepping-stone to new technology. In the late 2000s, things started opening up, and it is amazing what you see today.

NASA got involved in around 2010. It was after NSF had launched satellites. In a way, I’m glad NASA did not say I’m here to help you. They did not pay attention, and their biggest complaint was that it was too small for their instruments. Everybody thought that the CubeSat was a dumb idea. Now with small instruments being developed and with the support of NASA and NSF, CubeSats have evolved to become an important platform for space-based research.

6 The PocketQube

The main purpose of the CubeSat initiative was to educate students. One of the big factors in CubeSat missions is the launch price. When CubeSats became popular, the launch price went up because the big guys could afford to pay it. So in about 2005, I said I don’t like this, so I started thinking about ways to start making a little satellite and launch those in place of the CubeSat. So, I looked at the CubeSat design and said: “well if we slice this thing up, we can make eight PocketQubes in the space of a CubeSat in the PPOD.” I was trying to figure out how to put small satellites together to be like a CubeSat and how to separate them after they are deployed. I made model after model trying to figure out how to do it. The final design is a 5-cm cube and has a mass of 250 g (one PQU). When the PocketQube design was finalized, I set about trying to find a launch for it.

Beginning in 2009, I had moved to the Space Science Center at Morehead State University (MSU) where we worked with an organization called Kentucky Space to develop the PocketQube specifications. The team at Morehead State University, led by Ben Malphrus, had a long-term relationship with the Italian group from the University of Rome. I had introduced Ben to Fillipo Graziani, Dean of the Aerospace Engineering School and founder of the Group of Astrodynamics for the Use of Space Systems company (GAUSS) in around 2004. The GAUSS team had a relationship with the Russians and had flown several times on the Dnepr rockets. We worked out an arrangement to launch the first round of PocketQubes on a Dnepr launch. They were deployed from UniSat-5, a microsat built by the Italians, in much the same way that OPAL had deployed the first picosats. The first four PocketQube Satellites were launched on November 21, 2013. They included T-LogoQube/Eagle-1 (Morehead

State University and Sonoma State), QubeScout-S-1 University of Maryland, \$50Sat/Eagle-2 (An Amateur Radio group and Morehead State University), and WREN (Stadoko UG). T-LogoQube/Eagle-1 and \$50Sat were successful. The \$50Sat worked over 2 years, and T-LogoQube led by Garrett Jernigan launch worked for about 6 months. But one of the interesting technologies used was that the radio in both were RFM-22B that cost \$15, and they worked. A number of people said that these inexpensive radios would not work in space.

Since the successful launch of the first PocketQubes, many others have been in development. Workshops have been held at NASA Ames and at TU Delft (the Netherlands) and are now an annual event. At least 30 PocketQubes are in development. While most of them come from academia, several companies build PocketQubes, such as GAUSS Srl, Fossa Systems KSF Space Foundation, and Alba Orbital. PocketQube projects have even been the subject of Kickstarter campaigns and are popular with amateur radio satellite groups.

7 The ThinSat

In about 2015, I was at a SmallSat conference and attended a NASA Town Hall Meeting. I asked why we can't put CubeSats in the trunk of the Dragon capsule. They said it's all filled up with stuff, but if you want to put something on the second stage, there may be room. You will only get a short orbit life—maybe 4 or 5 days. I said I will take it! But I couldn't find anybody that would consider using this second stage for small satellite launches. The next year at the SmallSat conference I met a guy from Orbital Sciences Corporation, and I asked how about getting this on the second stage of your Antares? I had a PocketQube ready to launch. I had asked him for a launch for years, and he had laughed at me. This time he said that I might have something here; his name was Warren Frick. A few weeks later, he had talked to some people at Orbital ATK, and they said we might have a launch for you. Sometimes, things happen quickly!

We started looking at deployers. I found out that Orbital-ATK would prefer to use the Canisterized Satellite Dispenser (CSD) developed by Planetary Systems Corporation. I contacted Walter Holemans at Planetary Systems, and he met with me and Dale Nash from Virginia Space at an Orbital ATK meeting. They showed a manifest with PocketQubes. That was 1 month from the first time I met Warren Frick at the Utah Conference.

I then met with Hank Voss from Near Space Launch, whom I had known for many years. Hank was using a Globalstar radio that allowed you to send your data from your CubeSat to the Globalstar Network of satellites and then to the ground using their Internet network. So, it was not necessary to have your own ground station. The GlobalStar radio would fit in the little PocketQube. Using GlobalStar would solve the problem of communicating with large numbers of PocketQubes on a single launch.

Hank helped us design the next generation of PocketQubes. A couple of months later, Hank came back, and we had a meeting at Morehead State. Hank presented a different design for the PocketQube. Rather than a cube, he proposed satellites with the size of slices of a bread. Everybody talked about how PPOD is like a bread box.

Other people had also proposed a slice of bread satellite. It would allow us to eliminate all these little tedious circuit boards and use one big circuit board. We proposed this to Dale Nash at Virginia space and Orbital-ATK, and it was accepted. We decided that we wanted to tether them together and that got everyone excited. We thought about putting a stiff wire in between the ThinSats rather than using a loose tether. Ultimately, we came up with these folding panels and everything looked good.

We got our first launch of the ThinSats in April of 2019. The ThinSats, as we called them, didn't operate quite as we had expected. Depending on the altitude they were released at and the ballistic coefficient, everybody predicted they might last 5 days to 2 weeks. But they only lasted 1 day and a half. What we figured is that when they came out, they were spinning which did not leave them all face on which represents the minimum ballistic coefficient. We ended up launching 64 of them, and I think that out of the 64 we got data out of probably over half of them. So, the first generation of ThinSats was a success.

To me the value of the ThinSat program is in the STEM activity in the schools. It is very difficult to get a satellite launched, and you go through a lot of pain to do that. What is the education that you get if you build a satellite and don't launch it? To me, you get 95% of the value because you get most of the of the education. The idea that I'm going to launch something into space inspires students. So, we can work with the students and instill in them the idea that they don't necessarily have to launch a satellite to get all this education. And the learning that they achieve translates to a variety of areas beyond aerospace because of the diversity of the satellite systems.

8 CubeSats take off

The two things that made the CubeSat successful were the size and the containment. The small size made the launches affordable, and the containment relieved the fear of the launch provider that the CubeSats would damage the launch vehicle or the primary satellite. In the beginning the launch costs and containment did not do us any good in the United States because none of the US launchers would give us a ride. We talked to Lockheed Martin and Martin Marietta at the time about launching these things, and they said: "well if you give us a half a million dollars to study it, we will consider it." That's kind of funny now because Lockheed Martin has announced its own CubeSat program. Everybody that I got in contact with declined. Nobody would give us an opportunity except the Russians—they were hungry at the time.

This whole thing has become a worldwide industry. The mistake that I made is that I should have somehow patented the CubeSat so I could have gotten a dollar for each one that was launched. The initial intent was education for the students. Sending something to space is cool. I liked it because of all the technologies—you have the computers, you have the radio, and you have to talk with payloads; there are many challenges to solve.

When we first started working on satellites, I really got intrigued. With CubeSats, we have this little thing that we can get launched cheaper, and it doesn't cost a lot to

get it built, and we could do it quickly. That is another key thing; it doesn't take 20 years to build a satellite from inception to launch. That was the whole intent of it. Because of these reasons, I hoped that we would see a lot of them in the Universities, but I did not anticipate the commercial applications that would come along. They are overwhelming.

I never anticipated the commercial applications, but the thing that is kind of interesting is that the CubeSat really had some role in stimulating the space program. It has to be a factor in ushering in the new space era and a significant factor at that. I always thought it was the same as what Apple did with the computer industry. Apple started a whole generation believing that everybody can have and work with computers—you don't have to be a technical student or an engineer. If you look at the apps that are on a cell phone now, you never would have thought about doing something like that because people that create those apps are not necessarily the people that know the technology in the cell phones. I hope this does the same thing that is to spawn a whole new generation of people that are looking for apps on satellites, but more than that, expanding the application of the technology more widely.

I never envisioned that the commercial application would be what it is now. Was this the thing that will reinvigorate the space program? When we built the first CubeSats, everybody said we can't put our instrument in it because they are too little. But we came along at the right time—the miniaturization of the electronics for commercial applications like for the cell phone coincided at the time. If we had been earlier, it might not have worked. But now, miniaturized instruments have come a long way and can support interesting science and a variety of practical applications. And I can raise the TRL level of my technology so I can use it with other missions. That has really sparked a lot of activity. Who would have ever thought that you would send CubeSats to Mars or have constellations of hundreds of CubeSats monitoring the Earth? It has really been very satisfying from that standpoint, but I worry that the education end is getting shorted—we need to keep students involved in CubeSats because it was really designed for them.